

# PAPER\_458 — MUGE Final 7-System Canonical: 10-Term Resonance Acceleration Suite

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**Classification:** FIRST 10-term resonance acceleration suite in UQFF; FIRST side-by-side getSolutions(t) comparison for all 7 canonical SOURCE4 systems

**Author:** Daniel T. Murphy

**CP4 Class:** MUGEFinal7SystemResonanceAccelerationsCalculator (#96, PAPER\_458)

## Abstract

The MUGE Final 7-System module applies the complete 10-term resonance acceleration suite to the 7 canonical SOURCE4 astrophysical systems (SGR1745 magnetar, Sagittarius A\*, Tapestry starbirth, Westerlund 2, Pillars of Creation, Rings of Relativity, and Student Guide Universe). Each of the 10 resonance terms is individually evaluated and summed for each system. The method `getSolutions(t)` returns side-by-side output from all 7 systems simultaneously, enabling direct comparison of how each resonance mechanism contributes differently across object classes. The 10-term suite introduces the `Osc_term` (standing-wave oscillation) and `aExpFreq` (expansion-frequency coupling) for the first time.

## 2. The 10-Term Resonance Acceleration Suite — PAPER\_458

### 2.1 Term Listing

| #  | Term                      | Symbol                        | Formula                                                        |
|----|---------------------------|-------------------------------|----------------------------------------------------------------|
| 1  | THz hole coupling         | <code>a_THz</code>            | $c^3 / (GMr) \times f\_THz^2$                                  |
| 2  | Vacuum differential       | <code>a_vac_diff</code>       | $\_vac, [SCm] \times V^{(1/3)} - \_vac, [UA] \times V^{(1/3)}$ |
| 3  | Super-frequency           | <code>a_SuperFreq</code>      | $\Sigma A\_k \sin(2 f\_k t), k=1..5$                           |
| 4  | Aether resonance          | <code>a_AetherRes</code>      | $\_vac, \textcolor{red}{SCm} V\_sys^{(1/3)}$                   |
| 5  | Ug4 vacuum                | <code>Ug4_i</code>            | $U\_A \_vac (1 + [UA][SCm])$                                   |
| 6  | Quantum frequency         | <code>a_QuantumFreq</code>    | $\hbar \_q / (M c^2 r) \times c$                               |
| 7  | Aether frequency          | <code>a_AetherFreq</code>     | $f\_aether \times r \times [SCm]$                              |
| 8  | Fluid frequency           | <code>a_FluidFreq</code>      | $\_fluid \times f\_fluid^2 \times r$                           |
| 9  | Oscillation standing wave | <b><code>Osc_term</code></b>  | $A\_osc \cos(k\_osc r) \sin(\_osc t)$                          |
| 10 | Expansion frequency       | <b><code>a_ExpFreq</code></b> | $H\_0 \times c \times \sin(2 H\_0 t)$                          |

### 2.2 New Terms: `Osc_term` and `a_ExpFreq`

**`Osc_term` — Standing wave oscillation (FIRST in UQFF):**

$$Osc\_term(r, t) = A_{osc} \cos(k_{osc} r) \sin(\omega_{osc} t)$$

With  $A_{\text{osc}} = \text{system-dependent amplitude}$ ,  $k_{\text{osc}} = 2\pi/r_{\text{char}}$ ,  $\omega_{\text{osc}} = 2\pi f_{\text{char}}$ .

The  $\text{Osc\_term}$  represents **gravitational standing waves** in the system's characteristic cavity — analogous to the Schumann resonance for electromagnetic standing waves in the Earth-ionosphere cavity, but applied to the gravitational field.

**a\_ExpFreq — Expansion-frequency coupling (FIRST in UQFF):**

$$a_{\text{ExpFreq}}(t) = H_0 \cdot c \cdot \sin(2\pi H_0 t)$$

$$= 2.27 \times 10^{-18} \times 3 \times 10^8 \times \sin(2\pi \times 2.27 \times 10^{-18} t)$$

$$= 6.81 \times 10^{-10} \sin(1.427 \times 10^{-17} t) \text{ m/s}^2$$

Period:  $T_{\text{ExpFreq}} = 1/H_0 = 4.41 \times 10^{17} \text{ s} = 13.97 \text{ Gyr}$  (Hubble time). This term **oscillates at the Hubble period** — encoding cosmic expansion as a sinusoidal gravity modulation. At present epoch ( $t = t_{\text{H}}$ ),  $a_{\text{ExpFreq}} = 6.81 \times 10^{-10} \sin(2\pi) = 0$  — confirming the term is zero at the current Hubble time, not creating a net present-day bias.

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### 3. Full Resonance Equation

$$g_{\text{res}}^{(j)}(r, t) = g_{\text{Newton}}^{(j)}(1 + H_z t)(1 - B/B_{\text{crit}}) + \sum_{i=1}^{10} a_i^{(j)}(r, t)$$


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### 4. getSolutions(t) Results for 7 Canonical Systems

At  $t = 1 \text{ Gyr} = 3.156 \times 10^{17} \text{ s}$ :

#### 4.1 SGR 1745-2900 Magnetar

| Term               | Value (m/s <sup>2</sup> )                                                  |
|--------------------|----------------------------------------------------------------------------|
| g_Newton           | $3.716 \times 10^{12}$                                                     |
| a_THz              | $\sim 7.26 \times 10^2$                                                    |
| a_AetherRes        | $\sim 4.9 \times 10$                                                       |
| Osc_term           | $\sim 1 \times 10^{-3}$ (oscillatory)                                      |
| a_ExpFreq          | $\sim -6.81 \times 10^{-10} \sin(14.27) \quad 4.1 \times 10^{-10}$         |
| <b>g_res total</b> | <b><math>\sim 3.73 \times 10^{12}</math></b> (after UQFF coupling factors) |

#### 4.2 Sagittarius A\*

| Term     | Value (m/s <sup>2</sup> ) |
|----------|---------------------------|
| g_Newton | $\sim 6.25 \times 10^1$   |

| Term               | Value (m/s <sup>2</sup> )     |
|--------------------|-------------------------------|
| a_AetherFreq       | $\sim 1 \times 10^{-2}$       |
| a_FluidFreq        | $\sim 10^{-1}$                |
| a_ExpFreq          | $\sim 4.1 \times 10^{-1}$     |
| <b>g_res total</b> | <b><math>\sim 1.52</math></b> |

### 4.3 Tapestry Starbirth

| Term               | Value (m/s <sup>2</sup> )                    |
|--------------------|----------------------------------------------|
| g_Newton           | $\sim 2.65 \times 10^{-12}$                  |
| P_outflow          | $\sim 10^{-1}$                               |
| Osc_term           | $\sim 10^{-13}$                              |
| <b>g_res total</b> | <b><math>\sim 1.02 \times 10^{-1}</math></b> |

### 4.4 Universe Guide

| Term               | Value (m/s <sup>2</sup> )               |
|--------------------|-----------------------------------------|
| g_Newton           | $\sim 5.88 \times 10^{-1}$              |
| g_DM               | $\sim 1.58 \times 10^{-1}$              |
| a_ExpFreq          | $\sim 4.1 \times 10^{-1}$               |
| <b>g_res total</b> | <b><math>\sim 1.14 \times 10</math></b> |

## 5. Term Hierarchy Across 7 Systems

| Term        | Magnetar        | SgrA*  | Tapestry | Universe              |
|-------------|-----------------|--------|----------|-----------------------|
| a_THz       | <b>dominant</b> | small  | tiny     | tiny                  |
| a_AetherRes | large           | medium | small    | small                 |
| a_ExpFreq   | tiny            | tiny   | tiny     | <b>non-negligible</b> |
| Osc_term    | medium          | small  | medium   | small                 |
| a_vac_diff  | small           | small  | small    | negligible            |

**Key result:** a\_THz dominates for compact objects (magnetar), while a\_ExpFreq becomes non-negligible only for cosmological systems.

## 6. Standard Model Comparison

| Feature                    | SM   | UQFF PAPER_458             |
|----------------------------|------|----------------------------|
| Resonance terms in gravity | None | 10-term acceleration suite |

| Feature                   | SM             | UQFF PAPER_458                         |
|---------------------------|----------------|----------------------------------------|
| Hubble oscillation        | Not in gravity | $a\_ExpFreq = H c \sin(2 H t)$         |
| Standing-wave gravity     | Not in gravity | $Osc\_term = A \cos(k r) \sin(t)$      |
| Multi-system side-by-side | Separate codes | <code>getSolutions(t)</code> for all 7 |

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## 7. Testable Predictions

1. **a\_ExpFreq period = Hubble time:** At  $t=t_H$ ,  $a\_ExpFreq = 0$ . At  $t = t_H/4$ ,  $a\_ExpFreq$  is maximum. CMB power spectrum  $P(k)$  should show subtle periodic modulation with period corresponding to  $T\_ExpFreq = 1/H$ .
2. **Osc\_term cavity resonance:** For the magnetar ( $r = 10$  km cavity),  $Osc\_term$  at  $f\_char = c/(2r) = 1.5 \times 10^1$  Hz. Detectable as sub-millisecond periodic gravity wave from neutron star surface modes.
3. **a\_THz universality:** For all compact objects,  $a\_THz = c^3/(GMr) \times f\_THz^2$  — implies  $a\_THz/g\_Newton = (c/v\_escape)^2 \times (f\_THz r/c)^2$ , a universal ratio testable via GW observations.

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